THE Open GROUP

TOGAF® Standard – Architecture Deliverables Phase C

Version 1.0

October 2026

Barangay San Antonio Health Department Enterprise Architecture

Barangay San Antonio Health Center, Pasig City

Document Version History

Version Number	Version Date	Revised By	Description	Filename
1.0	October 22, 2026	UA&P Solution Architecture team	First issue, based on interviews (October 2–21, 2026) and verification answers (October 22, 2026)	C1 - Architecture Deliverables (Phase C).docx

Contents

1. Architectural Artifacts 4
2. Phase C: Data Architecture Artifacts 7
   - 2.1 Data Entity/Data Component Catalog 7
   - 2.2 Logical Data Diagram 8
   - 2.3 Data Entity/Business Function Matrix 8
   - 2.4 Data Security Diagram 9
   - 2.5 Data Dissemination Diagram 9
   - 2.6 Data Lifecycle Diagram 10
   - 2.7 Data Gap Analysis 10
3. Phase C: Application Architecture Artifacts 12
   - 3.1 Application Portfolio Catalog 12
   - 3.2 Interface Catalog 12
   - 3.3 Role/Application Matrix 12
   - 3.4 Application/Function Matrix 13
   - 3.5 Application Communication Diagram 13
   - 3.6 Application Gap Analysis 14

A Abbreviations 15

Tracking Information

Project Name	Barangay San Antonio Health Department Enterprise Architecture		
Prepared By	Rome Victor Mag-iba Sendin (Group Leader); Samuel Anthonius Badillo; Wilson Halim; Kevin Enrico Ang	**Document Version No.**	1.0
Title	Architecture Deliverables – Phase C: Information Systems Architecture	**Document Version Date**	October 22, 2026
Reviewed By	Dr. Glynnis Lynn U. Bonilla, Medical Officer IV (Executive Sponsor)	**Review Date**	October 23, 2026

Distribution List

From	Date	Phone/Email
Rome Victor Mag-iba Sendin, Group Leader, UA&P Solution Architecture	October 22, 2026	romevictor.sendin@uap.asia / 09456315746

To	Action*	Due Date	Phone//Email
Dr. Glynnis Lynn U. Bonilla, Medical Officer IV	Approve	October 23, 2026	bsahealthcenter@gmail.com
Hon. Thomas Raymond U. Lising, Punong Barangay	Review	October 23, 2026	bsahealthcenter@gmail.com
Hon. Rachel Marie S. Rustia-Olarte, Co-Chair, Committee on Health & Nutrition	Review	October 23, 2026	bsahealthcenter@gmail.com
Mr. Joel Pira, Instructor, Solution Architecture (UA&P)	Review	October 23, 2026	joel.pira@uap.asia

* Action Types: Approve, Review, Inform, File, Action Required, Attend Meeting, Other (please specify)

1. Architectural Artifacts

While the Enterprise Metamodel is used to support the structuring of architectural information, most stakeholders do not need or wish to know the detail contained within the Architecture Content Framework in this manner. Therefore, the use of catalogs, matrices, and diagrams is introduced to facilitate the presentation of architectural information so that it may be used for reference and governance purposes more readily.

Catalogs are lists of building blocks of a specific type or related types, matrices are grids that show relationships between two or more entities, and diagrams are graphical renderings of architectural content.

In summary, the result of an Architecture Development Method (ADM) developed architecture consists of a number of defined Architecture Building Blocks populated into architecture catalogs, with relationships specified between those building blocks in architecture matrices, and/or also presented as diagrams, as appropriate to satisfy the stakeholder concerns.

The TOGAF® Standard provides a set of recommended artifacts, summarized as follows:

ADM Phase	**Artifact**
Preliminary Phase	Principles catalog
Phase A	Stakeholder Map matrix Value Chain diagram Solution Concept diagram Business Model diagram Business Capability map Value Stream map

ADM Phase	Artifact
Phase B	Organization/Actor catalog Driver/Goal/Objective catalog Role catalog Business Service/Function catalog Location catalog Process/Event/Control/Product catalog Contract/Measure catalog Business Capabilities catalog Value Stream catalog Value Stream Stages catalog Business Interaction matrix Actor/Role matrix Value Stream/Capability matrix Strategy/Capability matrix Capability/Organization matrix Business Footprint diagram Business Service/Information diagram Functional Decomposition diagram Product Lifecycle diagram Goal/Objective/Service diagram Business Use-Case diagram Organization Decomposition diagram Process Flow diagram Event diagram Business Capability map (updated) Value Stream map (updated) Organization map
Phase C: Data Architecture	Data Entity/Data Component catalog Data Entity/Business Function matrix Application/Data matrix Conceptual Data Diagram Logical Data Diagram Data Dissemination diagram Data Security diagram Data Migration diagram Data Lifecycle diagram
Phase C: Application Architecture	Application Portfolio catalog Interface catalog Application/Organization matrix Role/Application matrix Application/Function matrix Application Interaction matrix Application Communication diagram Application and User Location diagram Application Use-Case diagram Enterprise Manageability diagram Process/Application Realization diagram Software Engineering diagram Application Migration diagram Software Distribution diagram

ADM Phase	Artifact
Phase D	Technology Standards catalog Technology Portfolio catalog Application/Technology matrix Environments and Locations diagram Platform Decomposition diagram Processing diagram Networked Computing/Hardware diagram Network and Communications diagram
Phase E	Project Context diagram Benefits diagram
Requirements Management	Requirements catalog

2. Phase C: Data Architecture Artifacts

2.1 Data Entity/Data Component Catalog

Entity	Key attributes	Where held today	Owner	Sensitivity / retention
Patient (ITR)	Record no., name, address, contact, birthdate, sex, civil status, PhilHealth no. and category, family/guardian, COVID doses	Paper ITR; City EMR	Encoder / nurse	Personal; 7 years
Consultation Encounter (Patient Chart)	Date, weight, height, BP, PR, RR, temp, O2 sat, LMP, sugar, complaint, history, exam, impression, plan, prescription, doctor, follow-up	Paper chart; City EMR	Doctor	Health; 7 years
Consent Record	Services checked, signature, authorized person, relationship, date/time	On ITR and every chart entry	Nurse	Legal; 7 years
Prescription	Medicine, dose, quantity, prescriber	Chart; EMR; YAKAP; dental eMED	Doctor / dentist	Health
GAMOT / YAKAP Entry	Beneficiary, address, date, UPSC, age, sex, medicines and quantity, follow-up date, QR code	PhilHealth portal; GAMOT logbook	Doctor / encoder	Health
Immunization Record	Vaccine, dose, date, brand	Chart; vaccine card; EMR	Nurse / midwife	Health
Maternal / Family Planning Record	Pregnancy, deliveries, FP method, unmet need	Chart; EMR; aides' tallies	Midwife	Sensitive
TB Case Record	Case, sputum / GeneXpert results, treatment	GeneXpert, sputum and TB logbooks; DOH ITIS	Nurse	Sensitive
Mental Health Record	Case, medication	Mental health logbook; locked	Nurse, midwife, doctor only	Highly sensitive
NCD / Maintenance Monitoring	Patient, medicine, quantity, date, BP	NCD form; monthly monitoring sheet	Nurse / aides	Health
Dental Record	Dental chart, vitals, treatment	Paper dental record; dental eMED	Dentist	Health
OPT Measurement	Child 0–59 months: name, birthdate, sex, weight, height, vitamin A, deworming	e-OPT form; NNC e-OPT tool; Google Sheets	BNS / nutritionist	Personal (children)
Referral	Patient, destination, reason, escort	Referral slip (no log)	Doctor / nurse	Health
Medicine Stock Transaction	Item, beginning balance, received, issued, ending balance	Manual utilization sheets; requisition and delivery slips; LMIS receipts	Nurse	Operational
Monthly Program Report	Program, period, tallies	Excel / Google Drive (personal); signed hard copies	Program owners	Aggregate
Ovitrap / Larvatrap Record	Household, result, date	Paper; 2-weekly report	Aides / BHWs	Aggregate
Equipment and Borrower Log	Item, borrower, date	Borrowers' and equipment logbooks	Custodian (Ms. Roa)	Operational
Visitor Log; Staff Attendance / DTR	Name, purpose, time	Logbooks; DTR	Security; each staff	Personal
Record Disposal Entry	Record type, location, disposal date, collector	Records storage/retention/disposal logbook	Custodian	Audit

2.2 Logical Data Diagram

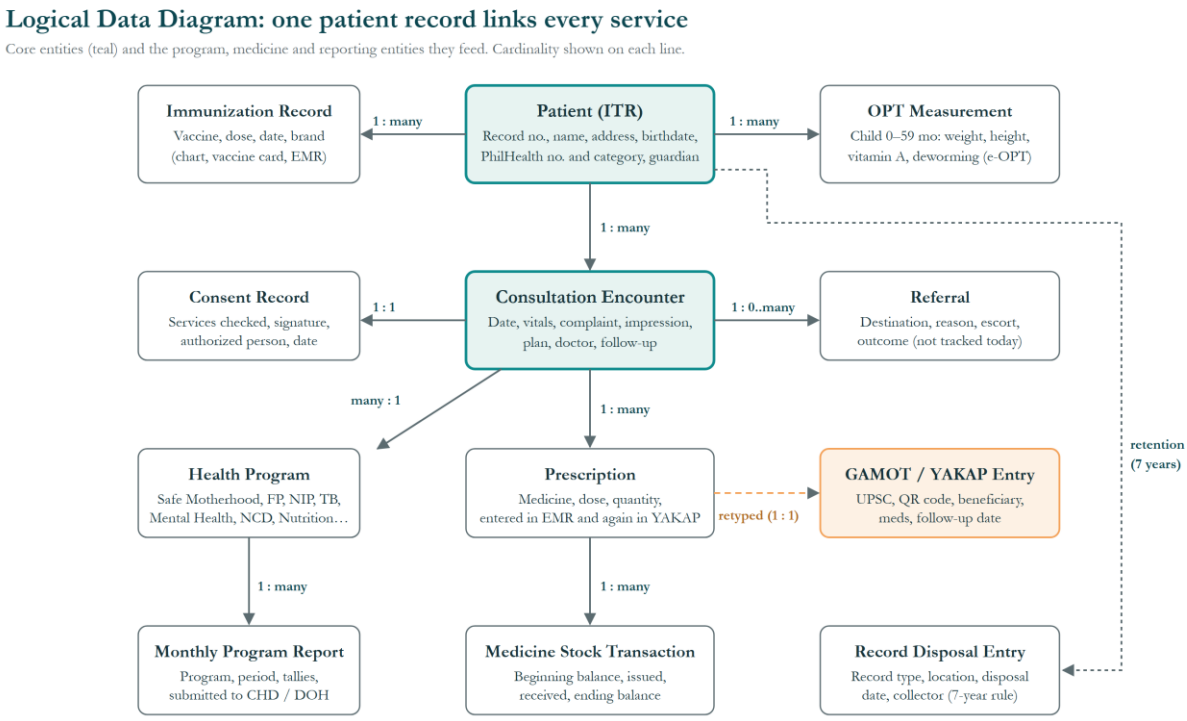


Figure 1. Logical Data Diagram.

How to read it: each box is a data entity with its key attributes; teal boxes are the core patient entities. Labels on the lines give the relationship: "1 : many" means one patient has many consultations; "1 : 0..many" means a patient may have none or several; "retyped (1 : 1)" marks the same data entered twice.

- **Patient (ITR) is the hub.** Its record number ties the paper chart to the EMR, applying "One Patient, One Record".
- **Prescription → GAMOT/YAKAP entry** is the main duplication: the same medicines entered twice.
- **Medicine Stock Transaction** connects dispensing to stock but lives on hand-written sheets.
- **Referral outcome** is not tracked today.
- **Record Disposal Entry** applies the DOH 7-year retention rule.
- Full attributes, owners and sensitivity are in the Data Entity/Data Component Catalog (Section 2.1).

2.3 Data Entity/Business Function Matrix

C = Create · R = Read · U = Update · D = Delete/dispose.

Entity	Registration	Consultation	Dispensing	Medicine logistics	Outreach	Reporting	Records custody
Patient (ITR)	C, U	R	R		R	R	D

Entity	Registration	Consultation	Dispensing	Medicine logistics	Outreach	Reporting	Records custody
Consultation Encounter	C (vitals)	C, U	R			R	D
Consent Record	C	C					D
Prescription		C	R	R		R	
GAMOT / YAKAP Entry		C	R			R	
Immunization Record		C			C	R	D
OPT Measurement					C, U	R	
Referral		C				R	
Medicine Stock Transaction			C	C, U		R	
Monthly Program Report						C, U	
Record Disposal Entry							C

2.4 Data Security Diagram

Entity	Who can access	Protection today	Gap
Paper patient records	All health center staff	Records room: only the main door is locked; custodian holds the key	No internal lock or key log
Mental health records	Nurse, midwife, doctor	Kept separate	Not documented as a rule
Nurse's program records	On request	Verbal permission; photocopy or photo given	No written access log
City EMR	Encoder, doctor, midwife, dentist (own accounts)	Individual accounts	Cross-barangay visibility unconfirmed (to ask Mr. Reyes)
Program files and chats	Owner of personal account	Personal passwords only	Work data on personal accounts
Consent	—	Signed on ITR and at every visit	Repeated per visit

2.5 Data Dissemination Diagram

Data	Sent to	How	Frequency
Program reports (NIP, TB, mental health, FP, maternal, NCD, medicines)	CHD program coordinators	Google Drive, then printed signed copies hand-carried	Monthly, quarterly, annual
Morbidity and service statistics	CHD Medstat; DOH FHSIS	EMR export	Monthly
Vaccine and mental-health medicine inventory	DOH	Inventory report	Monthly
TB cases	DOH	ITIS	Ongoing
OPT results	City Nutrition Office → region → national	NNC e-OPT	Jan–Mar
Nutrition reports (M1, BNS monthly, bi-annual)	City Nutrition Office	Google Drive	Monthly; June and December
Ovitrap results	Dengue Task Force office	Paper	Every 2 weeks

Data	Sent to	How	Frequency
YAKAP prescriptions and claims	PhilHealth; YAKAP pharmacy	Portal; QR code	Per patient
Equipment inventory	DOH, LGU, barangay	Custodian report	Monthly
Barangay health spending	City Accounting Office	Treasurer's report	Periodic

2.6 Data Lifecycle Diagram

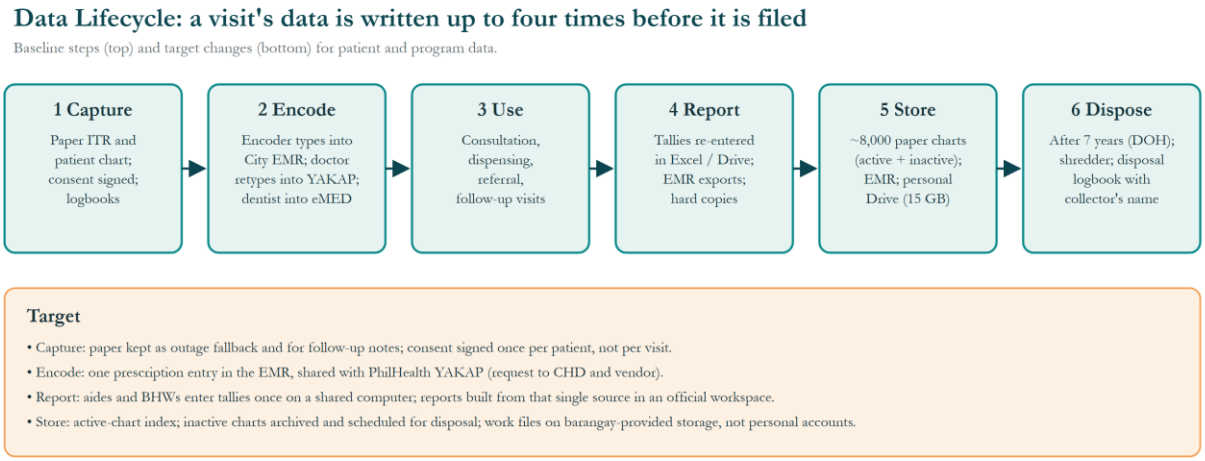


Figure 2. Data Lifecycle.

How to read it: the six numbered boxes are the stages a visit's data passes through, left to right; the "Target" panel below gives the planned change per stage.

- **Encode (stage 2)** and **Report (stage 4)** are where data is re-entered: encoder into the EMR, doctor into YAKAP, dentist into eMED, tallies into Excel/Drive.
- **Store (stage 5):** about 8,000 paper charts (active and inactive mixed) and personal 15 GB Drive accounts.
- **Dispose (stage 6)** is already well controlled: DOH 7-year rule, shredder, and a disposal logbook with the collector's name.
- **Targets:** consent once per patient (CHD approval); one prescription entry (City request); single tally source in an official workspace; active-chart index with inactive charts archived.

2.7 Data Gap Analysis

ID	Gap	Target	Controlled by
DG-1	Prescription data entered up to three times (chart, EMR, YAKAP / dental eMED)	Single entry shared across systems	City / vendor
DG-2	About 8,000 charts with active and inactive records mixed	Active-chart index; inactive charts archived on the 7-year schedule	Health center
DG-3	Program tallies captured on paper, then retyped	Tallies entered once on a shared computer	Health center / Barangay
DG-4	Work files on personal Google accounts (15 GB; nutritionist pays ₱200–400 a month)	Barangay-owned workspace and storage	Barangay

ID	Gap	Target	Controlled by
DG-5	Medicine stock tracked by hand	Digital stock record with reorder levels	Health center
DG-6	No referral log or outcome tracking (~30 referrals a month)	Referral register with outcomes	Health center
DG-7	Access to program records granted by verbal permission; records room locked only at the main door	Written access rule and key log; lockable cabinet	Health center
DG-8	Consent signed at every visit	Consent once per patient, renewed when services change	Health center (CHD approval)

3. Phase C: Application Architecture Artifacts

3.1 Application Portfolio Catalog

Application	Owner	Purpose	Users	Issues
City EMR (eMed)	City of Pasig; outsourced vendor (~2023)	Registration, vitals, diagnosis, prescriptions, report generation	Encoder, doctor, midwife, dentist	Slow at peak; internet-dependent; vendor-only changes
Dental eMED	City of Pasig	Dental charting and PhilHealth dental claims	Dentist, dental assistant	Separate from EMR
PhilHealth HCI Portal / YAKAP	PhilHealth	Member registration, YAKAP prescriptions, claims (own QR)	Doctor, encoder	Not linked to EMR
LMIS	City / DOH	Receipt of medicines, vaccines, nutrition commodities	Encoder	No daily dispensing view
ITIS	DOH	TB case management	Nurse	Separate entry
e-OPT Plus tool	National Nutrition Council	OPT Plus data entry and validation	BNS, nutritionist	Large files
Google Drive / Sheets	Personal accounts	Program reports, OPT master lists, CHD submissions	All staff	15 GB limit; no official domain
Messenger / Viber	Personal accounts	Team coordination (30+ groups)	All staff, Kapitan, Kagawad	Work and personal mixed
Facebook page	Health center	Public announcements	Staff	—

3.2 Interface Catalog

ID	From → To	Data	Method	Frequency
IF-1	Paper ITR/chart → City EMR	Patient, vitals, encounter	Manual retyping (encoder)	Every visit
IF-2	City EMR → PhilHealth YAKAP	Prescriptions	Manual retyping (doctor)	Every PhilHealth patient
IF-3	Dental eMED → PhilHealth	Dental claims	Manual retyping (dentist)	Every dental claim
IF-4	City EMR → CHD Medstat / DOH FHSIS	Morbidity and service statistics	Automatic export	Monthly
IF-5	Paper tallies → Google Sheets → CHD	Program figures	Manual; printed and hand-carried	Monthly
IF-6	Utilization sheets → LMIS / DOH inventory	Stock	Manual	Monthly
IF-7	OPT master list → e-OPT and Google Sheets	Child measurements	Manual, entered twice	Jan–Mar
IF-8	PhilHealth → YAKAP pharmacy	Prescription (QR)	Automatic via QR	Per patient

3.3 Role/Application Matrix

Role	City EMR	Dental eMED	PhilHealth	LMIS / ITIS	e-OPT	Google Drive	Chats
Doctor	X		X			X	X
Nurse	(not used)			ITIS		X	X
Midwife	X					X	X
Dentist and assistant	X	X	X				X
Encoder	X		X	LMIS		X	X
Nutritionist					X	X	X

Role	City EMR	Dental eMED	PhilHealth	LMIS / ITIS	e-OPT	Google Drive	Chats
BNS					X	X	X
Aides / BHWs							X

3.4 Application/Function Matrix

Function	City EMR	Dental eMED	PhilHealth	LMIS / ITIS	e-OPT	Google Drive	Chats	Paper
Registration	X		X					X
Consultation	X							X
Dental care		X	X					X
Dispensing	X		X					X
Medicine logistics				X			X	X
Outreach (OPT, MR-SIA)					X	X	X	X
Program reporting	X	X		X	X	X		X
Team communication							X	

3.5 Application Communication Diagram

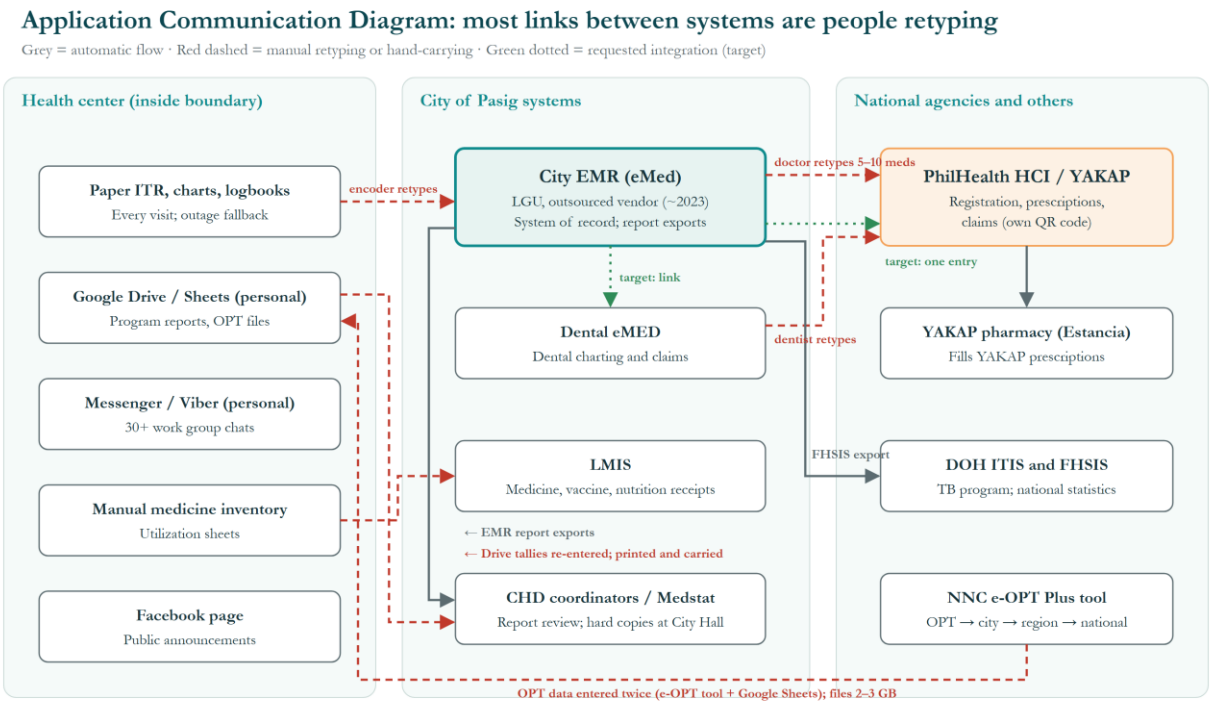


Figure 3. Application Communication Diagram.

How to read it: three zones: inside the health center (left), City of Pasig systems (middle), national agencies and others (right). Grey = automatic flow; red dashed = manual retyping or hand-carrying; green dotted = requested integration.

- **Automatic today:** EMR report exports (including to DOH FHSIS) and the PhilHealth QR code used by the YAKAP pharmacy.
- **Manual today:** encoder retypes paper charts into the EMR; doctor retypes 5–10 medicines into YAKAP; dentist retypes into PhilHealth; Drive tallies are re-entered, printed and carried to the CHD; OPT data is entered twice (e-OPT tool and Google Sheets; files of 2–3 GB).
- **Center-side tools** (Google Drive, Messenger and Viber with 30+ work chats) run on personal accounts.
- **Requested (WP8):** EMR → YAKAP single entry and a dental eMED link; both are City and vendor decisions.

3.6 Application Gap Analysis

ID	Gap	Candidate ABB	Controlled by
AG-1	City EMR and PhilHealth YAKAP not linked (doctor retypes 5–10 medicines)	Prescription interoperability	City / vendor / PhilHealth
AG-2	Dental eMED separate from EMR and PhilHealth	Dental record integration	City / vendor
AG-3	No tool for daily medicine stock and reorder alerts	Medicine inventory management	Health center
AG-4	No patient queuing tool (none in any Pasig health center)	Patient flow management	Health center (CHO approval)
AG-5	No official collaboration workspace	Shared work storage and messaging	Barangay
AG-6	Program tallies captured on paper and retyped	Single tally capture and report generation	Health center
AG-7	No appointment booking for dental and follow-ups	Appointment scheduling	Health center / City

A Abbreviations

Abbreviations, acronyms and reference codes used in this document and its figures, in alphabetical order.

Abbreviation	Meaning
ABB	Architecture Building Block (ABB-1 to ABB-9 are reference numbers)
ADM	Architecture Development Method, the step-by-step method of the TOGAF Standard
AG-1 …	Application gap reference number (Phase C)
BHW	Barangay Health Worker
BNS	Barangay Nutrition Scholar
BP	Blood pressure
CHD	City Health Department (Pasig City)
CHO	City Health Officer
DG-1 …	Data gap reference number (Phase C)
DOH	Department of Health
DTR	Daily Time Record
e-OPT	Electronic Operation Timbang Plus tool of the National Nutrition Council
eMed / eMED	Name of the City's electronic medical record system; “dental eMED” is its separate dental system
EMR	Electronic Medical Record
FHSIS	Field Health Services Information System (DOH)
FP	Family planning
GAMOT	PhilHealth Guaranteed and Accessible Medications for Outpatient Treatment
GB	Gigabyte (a unit of file size)
HCI	Health Care Institution (PhilHealth portal for health facilities)
Hon.	Honorable (title of elected officials)
IF-1 …	Interface reference number
ITIS	Integrated Tuberculosis Information System (DOH)
ITR	Individual Treatment Record (the patient's registration form)
LGU	Local Government Unit
LMIS	Logistics Management Information System
LMP	Last menstrual period
M1	FHSIS monthly report form
MR-SIA	Measles-Rubella Supplemental Immunization Activity
NCD	Non-communicable disease (e.g., hypertension, diabetes)
NIP	National Immunization Program
NNC	National Nutrition Council
O2 sat	Oxygen saturation
OPT / OPT Plus	Operation Timbang Plus: yearly weighing and measuring of children aged 0–59 months
PhilHealth	Philippine Health Insurance Corporation
PR	Pulse rate
QR	Quick Response (a scannable square barcode)
RR	Respiratory rate
TB	Tuberculosis
TOGAF	The Open Group Architecture Framework (the TOGAF® Standard)
UA&P	University of Asia and the Pacific
UPSC	Unique Prescription Security Code
WP1 … WP9	Work package (a group of related changes delivered together)

Abbreviation	Meaning
YAKAP	PhilHealth Yaman ng Kalusugan Program (primary-care benefit)